MIL-C-25973B(USAF)
18 January 1971
Superseding
MIL-C-25973A(USAF)
5 November 1963

MILITARY SPECIFICATION

* CONVERTER, LIQUID OXYGEN, GCU-12A/A

1. SCOPE

* 1.1 This specification covers one type of liquid oxygen converter with a capacitance gaging element. The converter is designated Type GCU-12A/A. It provides storage and pressurization of 5 liters of liquid oxygen to 70 pounds per square inch gage (psig), nominal pressure. The converter does not include warming coils.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-V-9050	Valves, Oxygen, Pressure Relief, Aircraft
MIL-C-25516	Connectors, Electrical, Miniature, Coaxial, Environment Resistant Type, General Specification for
MIL-C-25666	Converter, Liquid Oxygen, Capacitance Type Gaging, General Specification for

STANDARDS

Military

MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-831	Test Reports, Preparation of
MS33656	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal

FSC 1660

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DRAWING

Air Force

58J3533 Capacitance Chart - 5 Liter Liquid Oxygen Converter

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. Converters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.2 General specification. The requirements of MIL-C-25666 shall apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.3 Design. The converter shall conform to the dimensions and general design shown on figures 1, 2, 3, 4, and 5. The converter shall have a minimum capacity of 5 liters of liquid oxygen measured at 1 atmosphere and shall be designed for an operating pressure of 70 psig.

3.3.1 Schematic diagram. The converter shall be designed in accordance with the schematic diagram shown on figure 6.

3.3.2 Design strength. The filled converters shall be designed to withstand the following g forces without damage:

Vertical (down)	14g
Vertical (up)	6g
Lateral	6g

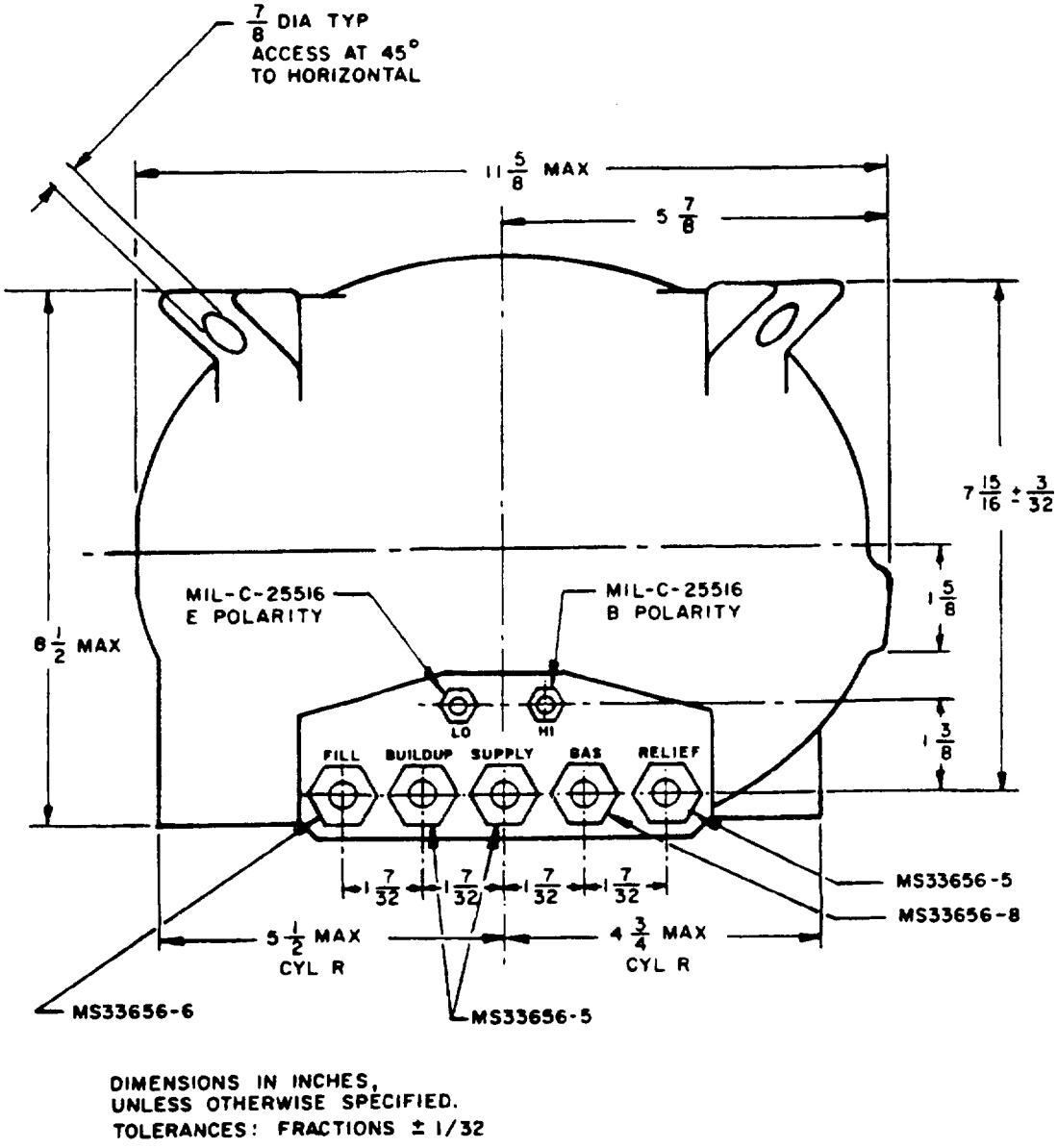
3.3.3 Relief valves. The relief valves shall conform to MIL-V-9050, type II.

3.3.4 Weight. The weight of the complete converter shall not exceed 12 pounds.

3.4 Performance

* 3.4.1 Permanent volumetric expansion. The converter shall meet the performance requirements for permanent volumetric expansion of MIL-C-25666 when subjected to a pneumatic or hydrostatic pressure of 120 psig.

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DIMENSIONS IN INCHES,
UNLESS OTHERWISE SPECIFIED.
TOLERANCES: FRACTIONS ± 1/32

FIGURE 1. Converter Configuration - Front View

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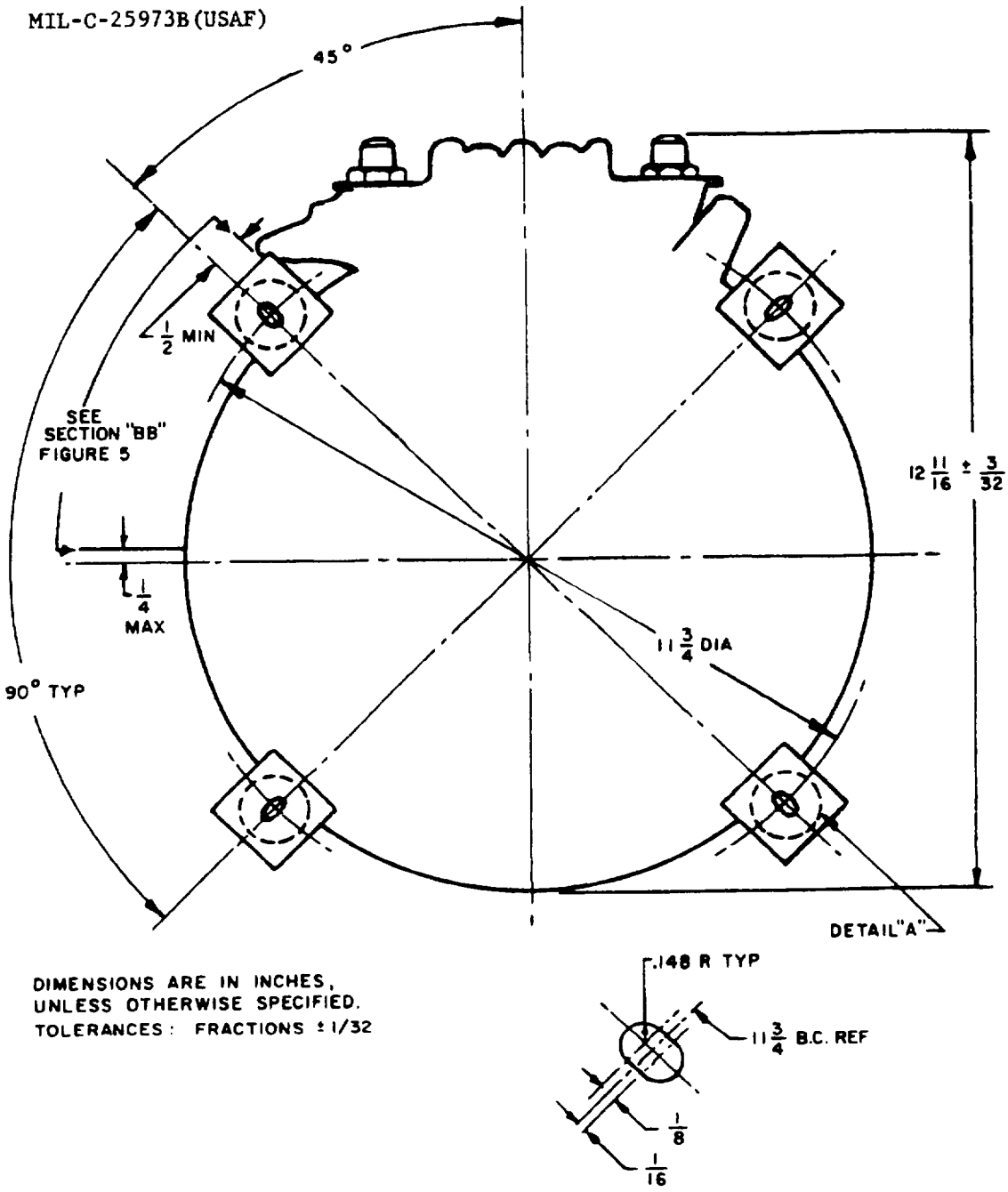


FIGURE 2. Converter Configuration - Top View

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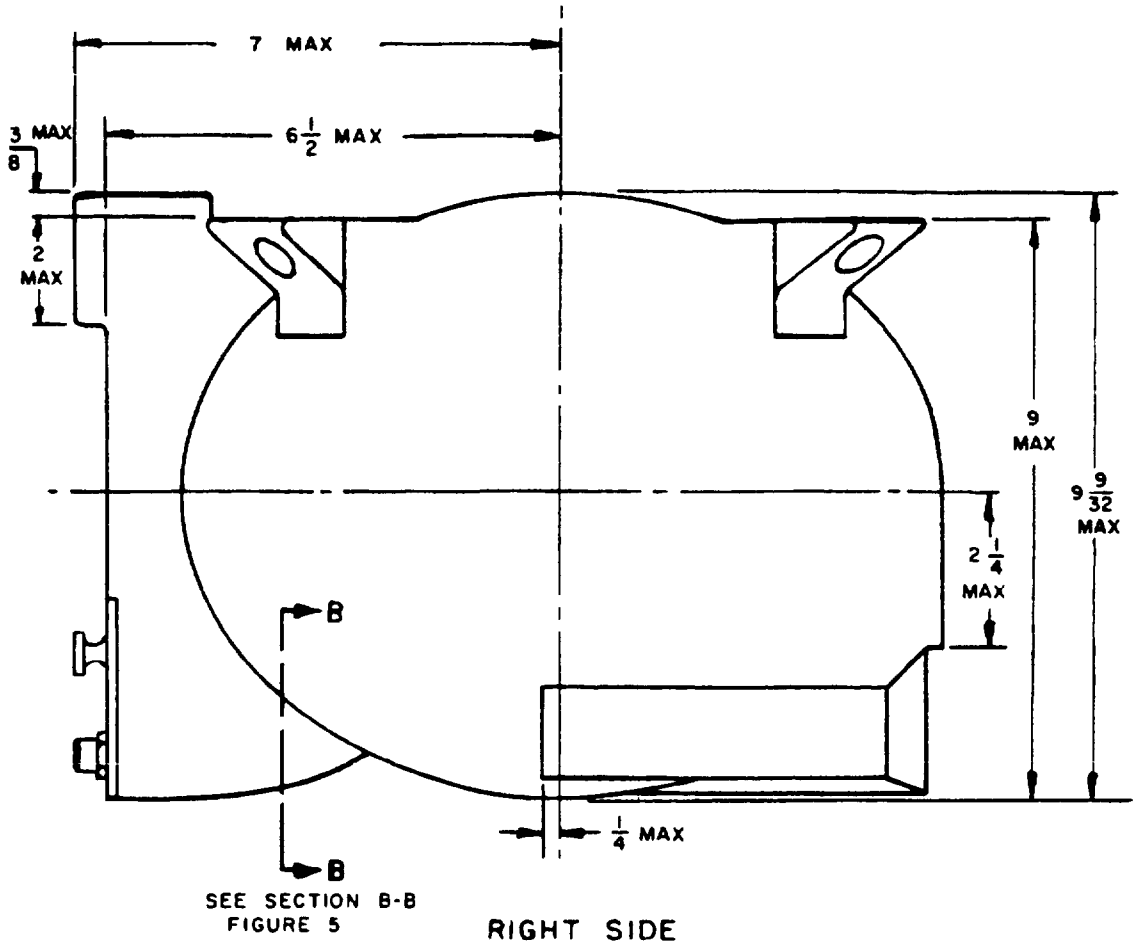
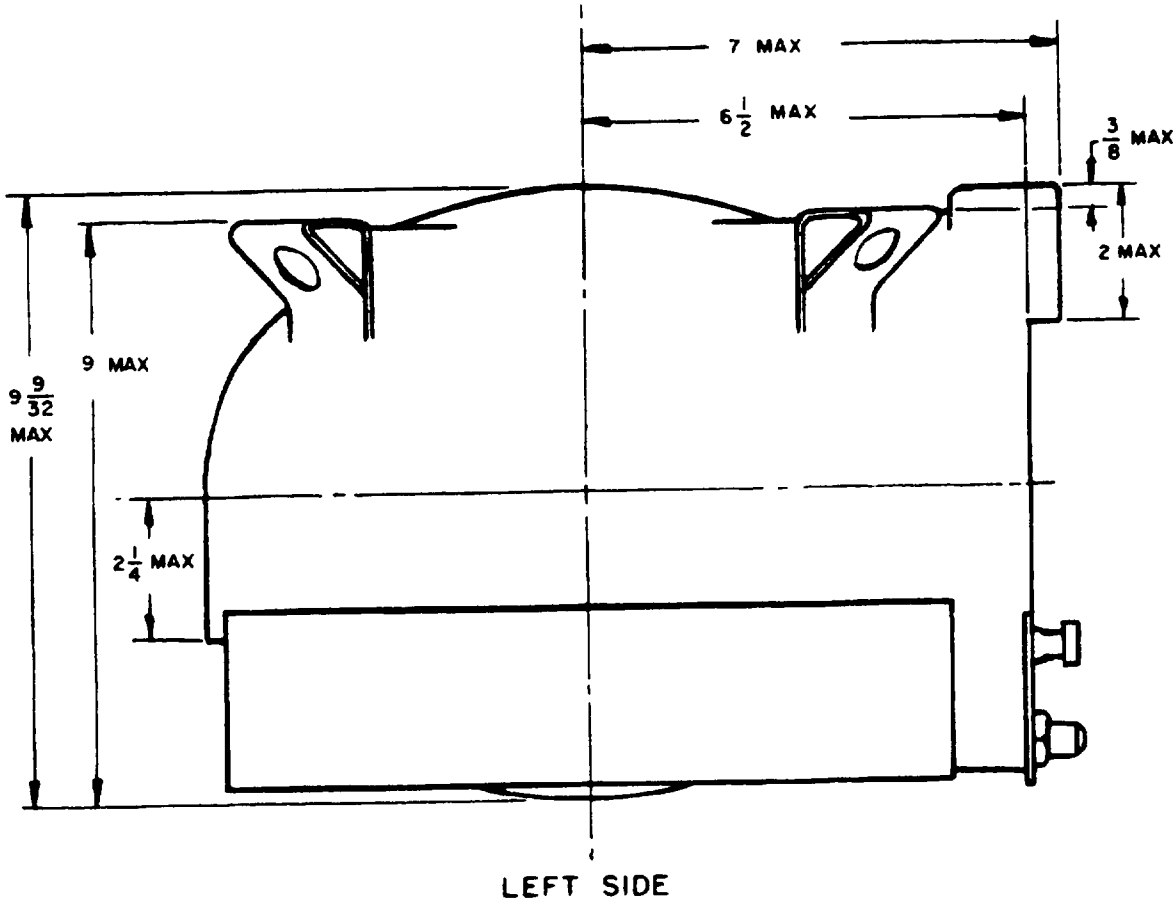


FIGURE 3. Converter Configuration - Right Side

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DIMENSIONS ARE IN INCHES,
UNLESS OTHERWISE SPECIFIED.
TOLERANCE ON FRACTIONS: ± 1/32

FIGURE 4. Converter Configuration - Left Side

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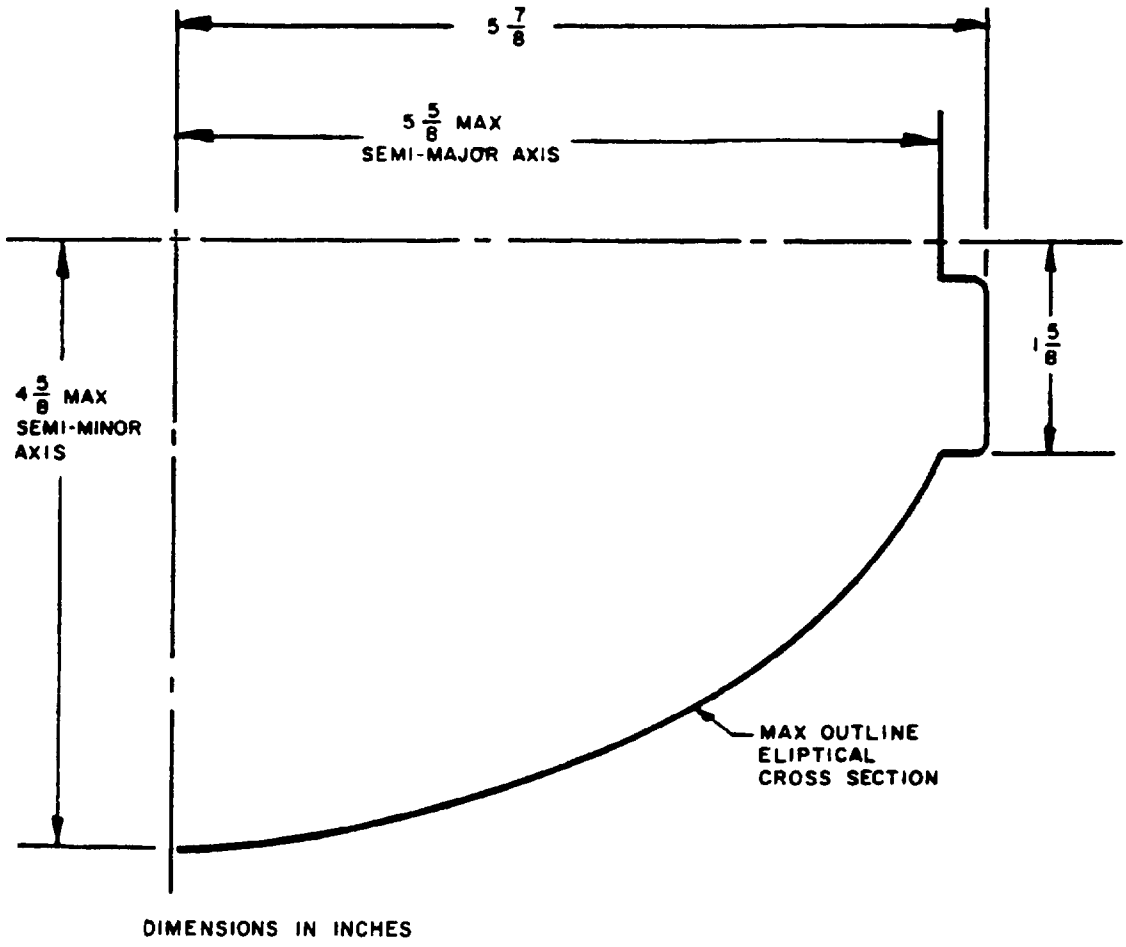


FIGURE 5. Converter Configuration - Section B-B

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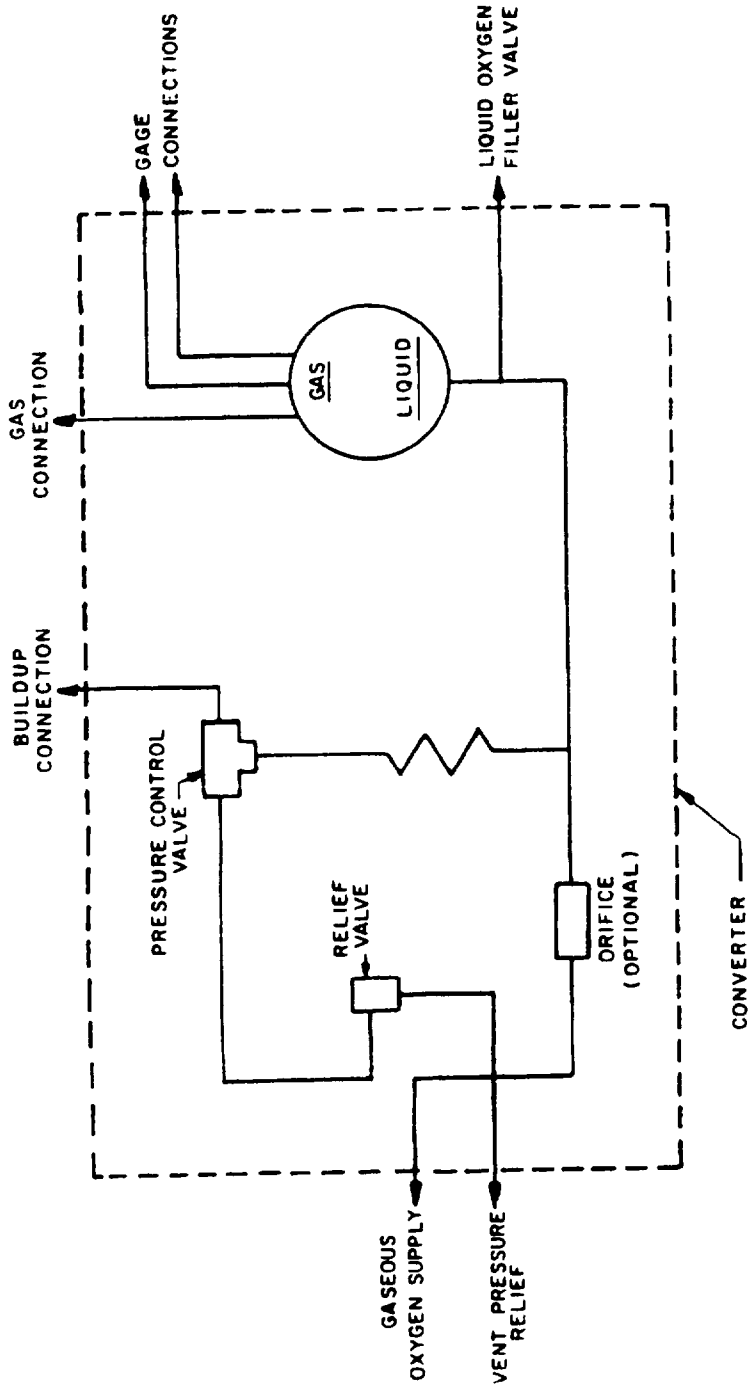


FIGURE 6. Schematic Diagram

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3.4.2 Evaporation loss. The evaporation loss of the converter shall not exceed 2.5 pounds of oxygen when the converter is tested as specified.

* 3.4.3 Final assembly pressure. The converter shall meet the performance requirements for the final assembly pressure test of MIL-C-25666 when subjected to a final assembly pressure of 85 psig.

3.4.4 Filling time. Filling time shall not exceed 5 minutes when the converter is tested as specified.

3.4.5 Buildup time. Buildup time shall not exceed 10 minutes when the converter is tested as specified.

3.4.6 Sensing element capacitance. The capacitance of the sensing element of the empty converter shall be 63.5 ±0.5 uuf.

3.4.7 Delivery rate. The converter shall deliver constant gas flows up to 72 liters of gaseous oxygen per minute for not less than 45 minutes with the operating pressure remaining between 70 and 80 psig after the first 15 minutes of operation.

3.4.8 Extreme temperature operation. The operating pressure shall remain between 70 and 85 psig after the first 15 minutes of flow at the low temperature and between 65 and 80 psig after the first 15 minutes of flow at the high temperature.

* 3.4.9 Vibration. The evaporation loss of the converter shall not exceed 2.5 pounds of oxygen after the converter is subjected to the vibration tests as specified.

* 3.4.10 Flow cycling. The converter shall deliver oxygen as specified in the flow cycling requirement of MIL-C-25666 for not less than 45 minutes when tested as specified.

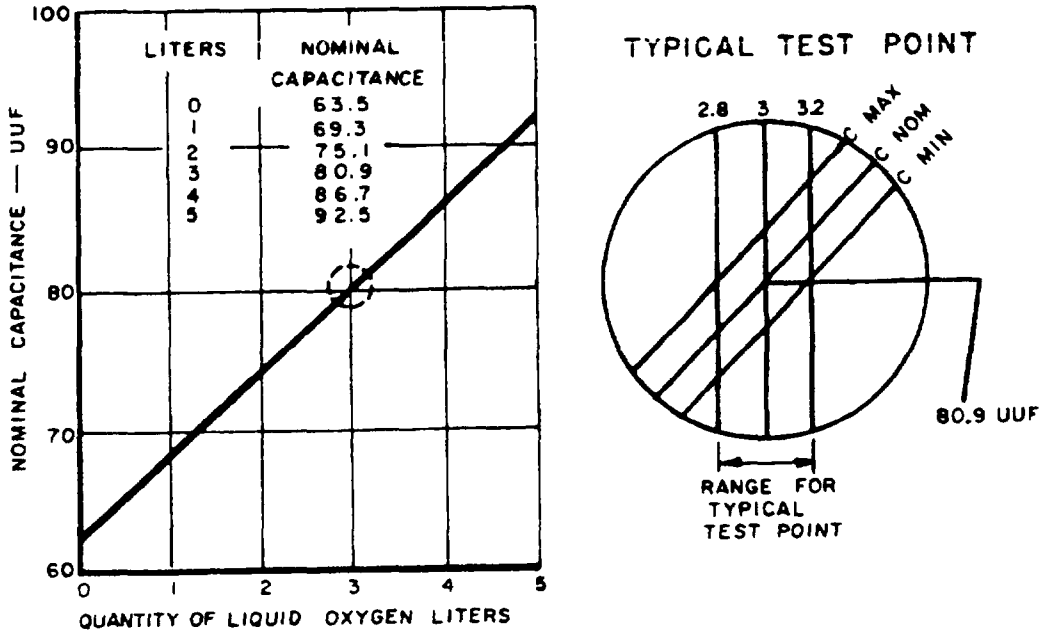
3.4.11 Sensing element calibration. The capacitance sensing element shall have characteristics which provide for a linear relationship between electrical capacitance and volume of liquid oxygen sensed. The relationship shall be in accordance with figure 7, using either the equations or the referenced drawing to determine tolerances. An example of use of the equations is as follows: At the 3-liter test point, suppose the actual quantity of liquid oxygen is 3.1 liters. The measured capacitance must be between C maximum (max) and C minimum (min).

C max - 5.9 x 3.1 + 64.1 = 82.4 micromicrofarad (uuf)

C min - 5.7 x 3.1 + 62.9 = 80.6 uuf

3.5 Marking of ports. Ports shall be marked as indicated on figure 1.

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CALCULATION OF MAXIMUM AND MINIMUM CAPACITANCES

FOR VALUES LESS THAN 1 LITER	FOR VALUES MORE THAN 1 LITER
C MAX = 6.0 L + 64.0	C MAX = 5.9 L + 64.1
C MIN = 5.6 L + 63.0	C MIN = 5.7 L + 62.9

C = CAPACITANCE IN UUF
L = EXACT QUANTITY OF LIQUID-OXYGEN AT TEST POINT IN LITERS

CALIBRATION LIMITS, BASED ON THE ABOVE EQUATIONS, ARE SHOWN IN DIRECT READING GRAPH FORM ON DRAWING 58J3533.

FIGURE 7. Calculation of Capacitances

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Testing, inspecting, and testing procedures. The testing, inspecting, and testing procedures shall be in accordance with MIL-C-25666 with the exceptions and additions specified herein.

* 4.3 Classification of tests. Inspecting and testing of the converters shall be classified as follows:

a. Qualification tests

b. Quality conformance tests.

4.4 Qualification testing

* 4.4.1 Qualification test samples tested by the manufacturer. The manufacturer shall use two converters for conducting the qualification tests specified in 4.4.4. One converter shall be subjected to the gunfire test in accordance with MIL-C-25666. The second unit shall be subjected to the remainder of the tests as required by MIL-C-25666.

* 4.4.2 Qualification test report. After the manufacturer completes the qualification tests, he shall prepare a qualification test report in accordance with MIL-STD-831 and shall furnish three complete copies of the report to the qualifying activity.

* 4.4.3 Qualification test sample for the qualifying activity. Along with the qualification test report, the manufacturer shall submit one converter. This sample may be a tested sample, except that it shall not be subjected to the test for gunfire specified in MIL-C-25666.

* 4.4.4 Qualification tests. The qualification tests shall consist of all the tests described under Test Methods of MIL-C-25666.

4.4.5 Qualification reliability tests. One qualification sample of the equipment shall be tested for reliability (using the flow cycling test) and shall be continued through a longevity test in accordance with MIL-STD-781.

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Acceptance shall be judged on the basis of the accept-reject criteria of test plan I of MIL-STD-781. Recording, data handling, and reporting procedures shall be in accordance with MIL-STD-781.

* 4.5 Quality conformance tests. The quality conformance tests shall consist of the Individual Tests and the Sampling Plans and Tests as specified in MIL-C-25666.

4.6 Test methods. The following shall apply to the applicable tests of MIL-C-25666.

4.6.1 Permanent volumetric expansion. The pneumatic or hydrostatic pressure used for this test shall be 120 psig.

4.6.2 Final assembly pressure. The pressure used for this test shall be 85 psig.

4.6.3 Vibration. The converter shall be vibrated in accordance with procedure XII of MIL-E-5272 except for the following:

a. The limiting acceleration shall be 4.5g.

b. The frequency range shall be from 5 to 50 hertz.

4.6.4 Flow cycling. The converter shall be subjected to the flow cycling test for not less than 45 minutes.

4.6.5 Reliability test level. Flow cycling testing shall be used for the reliability test.

5. PREPARATION FOR DELIVERY

5.1 The converter shall be prepared for delivery as specified in MIL-C-25666.

6. NOTES

6.1 Intended use. The converter covered by this specification is intended for use on F-104A and F-5A aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Selection of applicable levels of preservation and packaging and packing.

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* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Aeronautical Systems Division (ENCCE), Wright-Patterson Air Force Base, Ohio 45433, and information pertaining to qualification of products may be obtained from that activity.

* 6.4 The margins of this specification are marked with an asterisk to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Review activity:
Air Force - 71

Project No. 1660-F074

International interest (see MIL-C-25666)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 66 WHICH MAY BE USED.

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